# Riesz Representation Theorem

For any bounded linear functional $f$ on a Hilbert space $H$,
there exists a unique $y \in H$ such that

$$f(x) = \langle x, y \rangle \text{ for all } x \in H$$

- $\|f\| = \|y\|$ (norm preservation)
- The map $f \mapsto y$ is an isometric isomorphism $H^* \cong H$
- $H^* = H$ (dual space equals the space itself)

*Significance:*
*This theorem is fundamental to functional analysis.*
*It shows that Hilbert spaces are self-dual.*